ResearchAgent evidence report

Are Kestrel Semiconductor's advanced-packaging lead times in Malaysia getting shorter?

Synthetic demonstration: companies, publishers, numbers and exact passages are fictional. XRPL TESTNET · no real value.
Report: fixture · extractive-fixture
Fixture fallback: No LLM is configured; using validated final-answer extracts.
Research: fixture · extractive-fixture · Decision: fixture · metadata-fixture · Publisher: local
XRPL TESTNET · no real value

Budget S$2.00 · spent S$0.70 · refunded S$0.30

Executive answer

I run procurement for a mid-size fabless shop, and I have spent most of 2026 staring at Kestrel Semiconductor's Penang advanced-packaging queue. Here is the blunt version: the queue has been shrinking, and it has been shrinking faster than anyone on the buy side expected. I have the dated numbers, and I have the week counts, because that is the only way to talk about this business without getting lied to.

On 1 June 2026, Kestrel's Penang advanced-packaging lead time was 26 weeks. That was the number we built our H2 allocation plan around. It was ugly. It meant anything we booked in June landed in December, which meant our Q4 customer commitments were effectively locked before we had a single wafer out of the foundry. We were quoting our own customers at 30 weeks and padding the back end because we did not trust the 26.

By 1 August 2026, the same Penang line was quoting 22 weeks. That is a four-week improvement in eight weeks, which sounds modest until you annualise it. Then on 15 September 2026, the number came in at 18 weeks. So from 1 June 2026 to 15 September 2026, we went from 26 weeks to 18 weeks, a cut of 31%, while capacity grew from 11,000 units to 16,000 units a month. That is eight weeks of lead time gone in fifteen weeks of calendar time. I have been doing this for eleven years and I have never seen a packaging queue compress that fast without a demand shock behind it.

The obvious explanation is capacity. Kestrel commissioned Phase 2 at Penang on 15 August 2026. That is the date I have from the supplier's own capacity notice, and it lines up with the 1 August 2026 quote of 22 weeks, because the line was already loading Phase 2 tools in late July. Phase 2 is the advanced-packaging expansion, the one they have been teasing since 2024. It is real, it is running, and it is the reason the queue is moving.

A Kestrel statement dated 2 September 2026 says Phase 2 reached its rated capacity on 28 August 2026. The statement gives no unit figures. A second statement dated 20 September 2026 says the plant operated at 92 per cent of rated capacity in the week ending 18 September 2026. The record states no customer lead times.

On 29 September 2026 Kestrel Semiconductor announced a five-year agreement with TSMC for N3P

wafer capacity, 24,000 wafers a month from Q3 2027, plus advanced-packaging slots. The disclosure was made in an exchange filing published before the opening of trading. Kestrel will pay a US$1.1 billion prepayment and gave no margin guidance. The previous contract covered 16,000 wafers a month.

On July 25, 2026, Kestrel Semiconductor published its quarterly exchange filings with the Taiwan Stock Exchange and the Tokyo Stock Exchange. The filings, dated July 24, 2026, cover the period from April 1 to June 30, 2026. Kestrel reported net revenue of NT$142.3 billion for the quarter, a figure that appears in the consolidated income statement on page 12 of the Taiwan filing. The company also disclosed a gross margin of 48.7 percent, down from 51.2 percent in the same quarter of 2025. Operating income was NT$38.9 billion, and net income was NT$31.4 billion. These figures are stated in the filing without adjustment for foreign exchange fluctuations.

On 20 June 2026, Malaysia's Investment, Trade and Industry Ministry (MITI) published its monthly data-centre project register for May 2026. The register lists 14 approved data-centre projects with a combined committed investment of RM 9.4 billion. Of these, nine are located in Johor, three in Selangor, one in Penang and one in Negeri Sembilan. MITI states that the approvals cover a combined 412 megawatts of contracted electrical capacity.

Findings

1. I run procurement for a mid-size fabless shop, and I have spent most of 2026 staring at Kestrel Semiconductor's Penang advanced-packaging queue. Here is the blunt version: the queue has been shrinking, and it has been shrinking faster than anyone on the buy side expected. I have the dated numbers, and I have the week counts, because that is the only way to talk about this business without getting lied to. **CHALLENGES** [113]

2. On 1 June 2026, Kestrel's Penang advanced-packaging lead time was 26 weeks. That was the number we built our H2 allocation plan around. It was ugly. It meant anything we booked in June landed in December, which meant our Q4 customer commitments were effectively locked before we had a single wafer out of the foundry. We were quoting our own customers at 30 weeks and padding the back end because we did not trust the 26. **UNCERTAIN** [114]

3. By 1 August 2026, the same Penang line was quoting 22 weeks. That is a four-week improvement in eight weeks, which sounds modest until you annualise it. Then on 15 September 2026, the number came in at 18 weeks. So from 1 June 2026 to 15 September 2026, we went from 26 weeks to 18 weeks, a cut of 31%, while capacity grew from 11,000 units to 16,000 units a month. That is eight weeks of lead time gone in fifteen weeks of calendar time. I have been doing this for eleven years and I have never seen a packaging queue compress that fast without a demand shock behind it. **CHALLENGES** [115]

4. The obvious explanation is capacity. Kestrel commissioned Phase 2 at Penang on 15 August 2026. That is the date I have from the supplier's own capacity notice, and it lines up with the 1 August 2026 quote of 22 weeks, because the line was already loading Phase 2 tools in late July.

Phase 2 is the advanced-packaging expansion, the one they have been teasing since 2024. It is real, it is running, and it is the reason the queue is moving. **UNCERTAIN** [116]

5. A Kestrel statement dated 2 September 2026 says Phase 2 reached its rated capacity on 28 August 2026. The statement gives no unit figures. A second statement dated 20 September 2026 says the plant operated at 92 per cent of rated capacity in the week ending 18 September 2026. The record states no customer lead times. **UNCERTAIN** [4]

6. On 29 September 2026 Kestrel Semiconductor announced a five-year agreement with TSMC for N3P wafer capacity, 24,000 wafers a month from Q3 2027, plus advanced-packaging slots. The disclosure was made in an exchange filing published before the opening of trading. Kestrel will pay a US$1.1 billion prepayment and gave no margin guidance. The previous contract covered 16,000 wafers a month. **UNCERTAIN** [40]

7. On July 25, 2026, Kestrel Semiconductor published its quarterly exchange filings with the Taiwan Stock Exchange and the Tokyo Stock Exchange. The filings, dated July 24, 2026, cover the period from April 1 to June 30, 2026. Kestrel reported net revenue of NT$142.3 billion for the quarter, a figure that appears in the consolidated income statement on page 12 of the Taiwan filing. The company also disclosed a gross margin of 48.7 percent, down from 51.2 percent in the same quarter of 2025. Operating income was NT$38.9 billion, and net income was NT$31.4 billion. These figures are stated in the filing without adjustment for foreign exchange fluctuations. **UNCERTAIN** [54]

8. On 20 June 2026, Malaysia's Investment, Trade and Industry Ministry (MITI) published its monthly data-centre project register for May 2026. The register lists 14 approved data-centre projects with a combined committed investment of RM 9.4 billion. Of these, nine are located in Johor, three in Selangor, one in Penang and one in Negeri Sembilan. MITI states that the approvals cover a combined 412 megawatts of contracted electrical capacity. **UNCERTAIN** [98]

What purchases changed

The validated first and final findings differ; compare the cited answer versions below.

Impact: **QUALIFIES**

First answer · v1

1. A Kestrel statement dated 2 September 2026 says Phase 2 reached its rated capacity on 28 August 2026. The statement gives no unit figures. A second statement dated 20 September 2026 says the plant operated at 92 per cent of rated capacity in the week ending 18 September 2026. The record states no customer lead times. **UNCERTAIN** [4]

2. On 29 September 2026 Kestrel Semiconductor announced a five-year agreement with TSMC for N3P wafer capacity, 24,000 wafers a month from Q3 2027, plus advanced-packaging slots.

The disclosure was made in an exchange filing published before the opening of trading. Kestrel will pay a US$1.1 billion prepayment and gave no margin guidance. The previous contract covered 16,000 wafers a month. **UNCERTAIN** [40]

3. On July 25, 2026, Kestrel Semiconductor published its quarterly exchange filings with the Taiwan Stock Exchange and the Tokyo Stock Exchange. The filings, dated July 24, 2026, cover the period from April 1 to June 30, 2026. Kestrel reported net revenue of NT$142.3 billion for the quarter, a figure that appears in the consolidated income statement on page 12 of the Taiwan filing. The company also disclosed a gross margin of 48.7 percent, down from 51.2 percent in the same quarter of 2025. Operating income was NT$38.9 billion, and net income was NT$31.4 billion. These figures are stated in the filing without adjustment for foreign exchange fluctuations. **UNCERTAIN** [54]

4. On 20 June 2026, Malaysia's Investment, Trade and Industry Ministry (MITI) published its monthly data-centre project register for May 2026. The register lists 14 approved data-centre projects with a combined committed investment of RM 9.4 billion. Of these, nine are located in Johor, three in Selangor, one in Penang and one in Negeri Sembilan. MITI states that the approvals cover a combined 412 megawatts of contracted electrical capacity. **UNCERTAIN** [98]

5. The yen's recent move is a reminder that currency markets can turn quickly, especially when they are positioned one way. The rate differential remains the dominant driver, but it is not the only one. I will be watching the data, the rhetoric from the BoJ and the Fed, and the positioning of carry traders. As always, I will try to separate what is said and what gets done. That distinction has served me well in the past, and I suspect it will continue to do so. **CHALLENGES** [38]

6. Markets, as they often do, reacted swiftly. The yen strengthened initially, then gave back some gains as traders digested the details. The 10-year JGB yield ticked up by about 2 basis points, a modest move. But the real action was in the swaps market, where the implied path for the policy rate shifted. Before the meeting, overnight index swaps were pricing in a roughly 40 percent chance of a hike by the end of 2026. After the meeting, that probability rose to around 60 percent. This suggests that market participants interpreted the bank's language as slightly more hawkish than expected. However, I would caution against over-reading a single meeting. The bank has repeatedly stressed that it will be data-dependent, and the next few months of wage and price data will be crucial. **UNCERTAIN** [63]

7. This round allocated a little over 600 MW of new capacity across roughly two dozen buyers. Most contracts run for 21 years, which matches the economic life the developers assume for their panels. The buyers are a predictable mix: data-centre operators in Johor, two glove makers, a brewery, several electronics plants in Kulim and Perak, and one cement works that I suspect signed mainly for the press coverage. **UNCERTAIN** [69]

8. Take a hall with an IT load of 4.2 MW. Suppose the operators run a conventional chilled-water plant with a design PUE of 1.45. That gives a total facility load of 6.09 MW, of which 1.89 MW is

cooling and losses. Now suppose they decide to economise by admitting outside air at 32 degrees Celsius and 80 percent relative humidity, which is a typical afternoon in Kuala Lumpur or Jakarta. **UNCERTAIN** [80]

Final answer · v2

1. I run procurement for a mid-size fabless shop, and I have spent most of 2026 staring at Kestrel Semiconductor's Penang advanced-packaging queue. Here is the blunt version: the queue has been shrinking, and it has been shrinking faster than anyone on the buy side expected. I have the dated numbers, and I have the week counts, because that is the only way to talk about this business without getting lied to. **CHALLENGES** [113]

2. On 1 June 2026, Kestrel's Penang advanced-packaging lead time was 26 weeks. That was the number we built our H2 allocation plan around. It was ugly. It meant anything we booked in June landed in December, which meant our Q4 customer commitments were effectively locked before we had a single wafer out of the foundry. We were quoting our own customers at 30 weeks and padding the back end because we did not trust the 26. **UNCERTAIN** [114]

3. By 1 August 2026, the same Penang line was quoting 22 weeks. That is a four-week improvement in eight weeks, which sounds modest until you annualise it. Then on 15 September 2026, the number came in at 18 weeks. So from 1 June 2026 to 15 September 2026, we went from 26 weeks to 18 weeks, a cut of 31%, while capacity grew from 11,000 units to 16,000 units a month. That is eight weeks of lead time gone in fifteen weeks of calendar time. I have been doing this for eleven years and I have never seen a packaging queue compress that fast without a demand shock behind it. **CHALLENGES** [115]

4. The obvious explanation is capacity. Kestrel commissioned Phase 2 at Penang on 15 August 2026. That is the date I have from the supplier's own capacity notice, and it lines up with the 1 August 2026 quote of 22 weeks, because the line was already loading Phase 2 tools in late July. Phase 2 is the advanced-packaging expansion, the one they have been teasing since 2024. It is real, it is running, and it is the reason the queue is moving. **UNCERTAIN** [116]

5. A Kestrel statement dated 2 September 2026 says Phase 2 reached its rated capacity on 28 August 2026. The statement gives no unit figures. A second statement dated 20 September 2026 says the plant operated at 92 per cent of rated capacity in the week ending 18 September 2026. The record states no customer lead times. **UNCERTAIN** [4]

6. On 29 September 2026 Kestrel Semiconductor announced a five-year agreement with TSMC for N3P wafer capacity, 24,000 wafers a month from Q3 2027, plus advanced-packaging slots. The disclosure was made in an exchange filing published before the opening of trading. Kestrel will pay a US$1.1 billion prepayment and gave no margin guidance. The previous contract covered 16,000 wafers a month. **UNCERTAIN** [40]

7. On July 25, 2026, Kestrel Semiconductor published its quarterly exchange filings with the Taiwan Stock Exchange and the Tokyo Stock Exchange. The filings, dated July 24, 2026, cover

the period from April 1 to June 30, 2026. Kestrel reported net revenue of NT$142.3 billion for the quarter, a figure that appears in the consolidated income statement on page 12 of the Taiwan filing. The company also disclosed a gross margin of 48.7 percent, down from 51.2 percent in the same quarter of 2025. Operating income was NT$38.9 billion, and net income was NT$31.4 billion. These figures are stated in the filing without adjustment for foreign exchange fluctuations. **UNCERTAIN** [54]

8. On 20 June 2026, Malaysia's Investment, Trade and Industry Ministry (MITI) published its monthly data-centre project register for May 2026. The register lists 14 approved data-centre projects with a combined committed investment of RM 9.4 billion. Of these, nine are located in Johor, three in Selangor, one in Penang and one in Negeri Sembilan. MITI states that the approvals cover a combined 412 megawatts of contracted electrical capacity. **UNCERTAIN** [98]

Open questions

No open questions recorded.

Method

Report findings validated with the shared citation resolver against accessible exact passages. Decisions, receipts and sources copied from the persisted run snapshot. Conclusions shown as validated claim extracts.

Appendix · decisions and receipts

Ledger data from the persisted run; the writing model does not create these records.

value = gapMaterial × addressesGap × P(original) × (0.5 + 0.25 × credibility). Policy code enforces cap, budget and eligibility.

Round 1 · fixture · metadata-fixture

Gap: No accessible dated figures for lead times in weeks, or their trend. · material probability 0.900 · threshold 0.200

Candidate / price	Gap / original / credibility	Value / per S$	Verdict
Singapore's Monetary Policy: A Case Study in Rate Lag kopi-singapore-s-monetary-policy-a-case-study-in-rate-lag · S$0.10	0.010 / 0.900 / 1.000	0.005 / 0.047	SKIP_LOW_VALUE Value is below the model threshold.
Kestrel's Penang packaging lead times, week by week alphaleak-kestrel-penang-lead-times · S$0.30	0.960 / 0.900 / 1.000	0.467 / 1.555	BUY · would buy Clears the value threshold and spending policy.
Malaysia Renewable Energy Integration into Grid: Challenges notfinancialtimes-malaysia-renewable-energy-integration-into-grid-challenges · S$0.90	0.025 / 0.900 / 1.500	0.014 / 0.015	SKIP_LOW_VALUE Value is below the model threshold.
Kestrel Semiconductor advanced-packaging lead times in Malaysia: the dated series fab-floor-kestrel-penang-lead-times · S$0.40	0.739 / 0.900 / 1.000	0.359 / 0.898	BUY · would buy Clears the value threshold and spending policy.
Malaysia Grid Connection Queues – Status & Impact marketpulse-malaysia-grid-connection-queues-status-impact · S$0.20	0.020 / 0.020 / 1.000	0.000 / 0.001	SKIP_REWRITE Already acquired or a rewrite of existing evidence.
Interconnector Capacity Between Malaysia and Singapore: 2026 Outlook load-interconnector-capacity-between-malaysia-and-singapore-2026- · S$0.60	0.023 / 0.900 / 1.000	0.011 / 0.018	SKIP_LOW_VALUE Value is below the model threshold.
Advanced packaging lead times in early 2026: a reality check the-2026-q1-advanced-packaging-lead-times-a-reality-check · S$0.25	0.264 / 0.900 / 1.000	0.128 / 0.514	SKIP_LOW_VALUE Value is below the model threshold.

Candidate / price	Gap / original / credibility	Value / per S$	Verdict
Kestrel's TSMC deal: analysts trim margins mp-kestrel-deal-digest · S$0.20	0.024 / 0.020 / 1.000	0.000 / 0.001	SKIP_REWRITE Already acquired or a rewrite of existing evidence.

Round 2 · fixture · metadata-fixture

Gap: No accessible dated figures for lead times in weeks, or their trend. · material probability 0.900 · threshold 0.200

Candidate / price	Gap / original / credibility	Value / per S$	Verdict
Singapore's Monetary Policy: A Case Study in Rate Lag kopi-singapore-s-monetary-policy-a-case-study-in-rate-lag · S$0.10	0.010 / 0.900 / 1.000	0.005 / 0.047	SKIP_LOW_VALUE Value is below the model threshold.
Kestrel's Penang packaging lead times, week by week alphaleak-kestrel-penang-lead-times · S$0.30	0.960 / 0.900 / 1.000	0.018 / 0.061	SKIP_LOW_TRUST Publisher quarantined: never bought or cited.
Malaysia Renewable Energy Integration into Grid: Challenges notfinancialtimes-malaysia-renewable-energy-integration-into-grid-challenges · S$0.90	0.025 / 0.900 / 1.500	0.014 / 0.015	SKIP_LOW_VALUE Value is below the model threshold.
Kestrel Semiconductor advanced-packaging lead times in Malaysia: the dated series fab-floor-kestrel-penang-lead-times · S$0.40	0.739 / 0.900 / 1.000	0.359 / 0.898	BUY · would buy Clears the value threshold and spending policy.
Malaysia Grid Connection Queues – Status & Impact marketpulse-malaysia-grid-connection-queues-status-impact · S$0.20	0.020 / 0.020 / 1.000	0.000 / 0.001	SKIP_REWRITE Already acquired or a rewrite of existing evidence.
Interconnector Capacity Between Malaysia and Singapore: 2026 Outlook load-interconnector-capacity-between-malaysia-and-singapore-2026- · S$0.60	0.023 / 0.900 / 1.000	0.011 / 0.018	SKIP_LOW_VALUE Value is below the model threshold.
Advanced packaging lead times in early 2026: a reality check the-2026-q1-advanced-packaging-lead-times-a-reality-check · S$0.25	0.264 / 0.900 / 1.000	0.128 / 0.514	SKIP_LOW_VALUE Value is below the model threshold.
Kestrel's TSMC deal: analysts trim margins mp-kestrel-deal-digest · S$0.20	0.024 / 0.020 / 1.000	0.000 / 0.001	SKIP_REWRITE Already acquired or a rewrite of existing evidence.

Settlement receipts · XRPL TESTNET · no real value

Receipt / intent	Resource / version	Amount / settled at
x402-AD89391DD0F26CC910D60FE530B025CC7D547FE3AFDD5AD6FACD7FBCBD0D54F0 7452f60e-6c75-4011-a6ce-86c00e0bb698:1:alphaleak-kestrel-penang-lead-times:v1	alphaleak-kestrel-penang-lead-times · v1	S$0.30 = 0.03 XRP 2026-11-03T03:44:20.491Z XRPL tx AD89391DD0F26CC9… · ledger 22027608 rwi1i3TJfZKmLWoAiNkXUBZmdWVGFxBNRQ → rGeN8NcEVRVvWeFoyY9Df1TyiLq9AhoX5T
x402-3081E9C26915BDEA7A4FB9154B27C6E1C9F597121BAEE5F6967D8C63CD256365 7452f60e-6c75-4011-a6ce-86c00e0bb698:2:fab-floor-kestrel-penang-lead-times:v1	fab-floor-kestrel-penang-lead-times · v1	S$0.40 = 0.04 XRP 2026-11-03T03:44:33.248Z XRPL tx 3081E9C26915BDEA… · ledger 22027612 rwi1i3TJfZKmLWoAiNkXUBZmdWVGFxBNRQ → rnkdwreaQygabFAALPboYfDGtMXCHYoMAF
REFUND · failed proof 7452f60e-6c75-4011-a6ce-86c00e0bb698:1:alphaleak-kestrel-penang-lead-times:v1	alphaleak-kestrel-penang-lead-times · v1	−S$0.30 refunded by the writer tx B3765171BB6B2800…

Refunded S$0.30 · the refunded source is quarantined and never cited.

Source appendix · exact synthetic passages

Malaysia advanced-packaging output and capacity, Q3 2026

Open Records · or-penang-packaging-output-2026-q3 · v1 · free
Receipt IDs: none
Attribution: Open Records

[1] Exact synthetic passage · span p1 · open on the writer's site

> This record covers advanced-packaging output and capacity in Malaysia for the third quarter of 2026. It is compiled from company filings, ministry statements and trade-association releases. Figures are as reported; no adjustments have been made.

[2] Exact synthetic passage · span p2 · open on the writer's site

> Kestrel Semiconductor Sdn Bhd operates two advanced-packaging plants in Penang. The Phase 2 plant was commissioned on 15 August 2026, raising capacity from 11,000 to 16,000 advanced-packaging units a month. The company's filing states that Phase 1 continues at 11,000 units a month and that the two lines are not interchangeable. The record states no customer lead times.

[3] Exact synthetic passage · span p3 · open on the writer's site

> Kestrel's Q3 2026 output filing, submitted to the Malaysian Investment Development Authority on 12 September 2026, reports 38,400 units shipped for the quarter across both plants. The filing attributes 21,600 units to Phase 1 and 16,800 units to Phase 2. Phase 2 output is recorded from 15 August 2026 onward. The company lists no export destinations in the filing.

[4] Exact synthetic passage · span p4 · open on the writer's site

> A Kestrel statement dated 2 September 2026 says Phase 2 reached its rated capacity on 28 August 2026. The statement gives no unit figures. A second statement dated 20 September 2026 says the plant operated at 92 per cent of rated capacity in the week ending 18 September 2026. The record states no customer lead times.

[5] Exact synthetic passage · span p5 · open on the writer's site

> Three other operators file advanced-packaging figures for Malaysia. Sunline Assembly (M) Sdn Bhd reports 9,200 units a month at its Bayan Lepas plant, unchanged from Q2 2026. Its Q3 2026 output filing records 27,600 units shipped. The filing is dated 8 September 2026.

[6] Exact synthetic passage · span p6 · open on the writer's site

> Northgate Micro Sdn Bhd reports 7,500 units a month at its Kulim plant, unchanged from Q2 2026. Its Q3 2026 output filing records 22,500 units shipped. The filing is dated 9 September 2026.

[7] Exact synthetic passage · span p7 · open on the writer's site

> Aurex Packaging (M) Sdn Bhd reports 5,400 units a month at its Prai plant, unchanged from Q2 2026. Its Q3 2026 output filing records 16,200 units shipped. The filing is dated 10 September 2026.

[8] Exact synthetic passage · span p8 · open on the writer's site

> Combined reported capacity for the four operators is 38,100 units a month at the end of Q3 2026, against 33,100 units a month at the end of Q2 2026. Combined reported Q3 2026 output is 104,700 units. The record states no customer lead times.

[9] Exact synthetic passage · span p9 · open on the writer's site

> Two operators file advanced-packaging figures outside Penang and Kulim. Straits Test & Assembly Sdn Bhd reports 4,800 units a month at its Johor Bahru plant, unchanged from Q2 2026. Its Q3 2026 output filing records 14,400 units shipped. The filing is dated 11 September 2026.

[10] Exact synthetic passage · span p10 · open on the writer's site

> Borneo Micro Assembly Sdn Bhd reports 2,600 units a month at its Kuching plant, unchanged from Q2 2026. Its Q3 2026 output filing records 7,800 units shipped. The filing is dated 11 September 2026.

[11] Exact synthetic passage · span p11 · open on the writer's site

> Combined reported capacity for the two operators is 7,400 units a month at the end of Q3 2026, unchanged from the end of Q2 2026. Combined reported Q3 2026 output is 22,200 units. The record states no customer lead times.

[12] Exact synthetic passage · span p12 · open on the writer's site

> The Malaysia Semiconductor Industry Association released its Q3 2026 packaging summary on 25 September 2026. The summary lists six reporting members and gives a combined capacity of 45,500 units a month at the end of Q3 2026. It gives a combined output of 126,900 units for the quarter. The association notes that two members did not file Q3 2026 figures.

[13] Exact synthetic passage · span p13 · open on the writer's site

> The Ministry of Investment, Trade and Industry published an update on 18 September 2026 covering approved advanced-packaging projects. The update lists four approved projects with a combined planned capacity of 21,000 units a month. It gives no commissioning dates. The update states that approved capacity is not the same as installed capacity.

[14] Exact synthetic passage · span p14 · open on the writer's site

> The Malaysian Investment Development Authority published Q3 2026 filing deadlines on 1 July 2026. The deadlines are 10 September 2026 for Penang and Kulim operators and 11 September 2026 for all other operators. All filings cited in this record were submitted by the stated deadlines.

[15] Exact synthetic passage · span p15 · open on the writer's site

> At the end of Q2 2026, reported capacity for the six operators covered by the association summary was 40,500 units a month. At the end of Q3 2026 it was 45,500 units a month. The change is 5,000 units a month. Kestrel's Phase 2 plant accounts for the whole of the change.

[16] Exact synthetic passage · span p16 · open on the writer's site

> Reported Q2 2026 output for the same six operators was 112,200 units. Reported Q3 2026 output was 126,900 units. The change is 14,700 units.

[17] Exact synthetic passage · span p17 · open on the writer's site

> No operator filing cited in this record gives customer lead times. The record states no customer lead times. No operator filing cited in this record gives utilisation rates other than Kestrel's 92 per cent figure for the week ending 18 September 2026. No operator filing cited in this record gives export destinations.

[18] Exact synthetic passage · span p18 · open on the writer's site

> Currency figures are not reported in the filings cited. Headcount figures are not reported in the filings cited. The record does not include figures for operators that did not file.

[19] Exact synthetic passage · span p19 · open on the writer's site

> Kestrel Semiconductor Sdn Bhd: Q3 2026 output filing, 12 September 2026; commissioning statement, 2 September 2026; capacity statement, 20 September 2026.

[20] Exact synthetic passage · span p20 · open on the writer's site

> Sunline Assembly (M) Sdn Bhd: Q3 2026 output filing, 8 September 2026.

[21] Exact synthetic passage · span p21 · open on the writer's site

> Northgate Micro Sdn Bhd: Q3 2026 output filing, 9 September 2026.

[22] Exact synthetic passage · span p22 · open on the writer's site

> Aurex Packaging (M) Sdn Bhd: Q3 2026 output filing, 10 September 2026.

[23] Exact synthetic passage · span p23 · open on the writer's site

> Straits Test & Assembly Sdn Bhd: Q3 2026 output filing, 11 September 2026.

[24] Exact synthetic passage · span p24 · open on the writer's site

> Borneo Micro Assembly Sdn Bhd: Q3 2026 output filing, 11 September 2026.

[25] Exact synthetic passage · span p25 · open on the writer's site

> Malaysia Semiconductor Industry Association: Q3 2026 packaging summary, 25 September 2026.

[26] Exact synthetic passage · span p26 · open on the writer's site

> Ministry of Investment, Trade and Industry: approved projects update, 18 September 2026.

[27] Exact synthetic passage · span p27 · open on the writer's site

> Malaysian Investment Development Authority: Q3 2026 filing deadlines, 1 July 2026.

[28] Exact synthetic passage · span p28 · open on the writer's site

> This record is dated 30 September 2026. It will be updated if further filings are received.

Yen Strength and the Rate Differential: Implications for Investors

Basis Points · basis-yen-strength-and-the-rate-differential-implications-for-inve · v1 · free
Receipt IDs: none
Attribution: Basis Points

[29] Exact synthetic passage · span p1 · open on the writer's site

> Speaking as an ex-central banker, I've learned to be wary of drawing firm conclusions from short stretches of market action. The yen has strengthened against the dollar in recent weeks, and that move has prompted a familiar debate about whether the long-standing rate differential between Japan and the United States is finally narrowing in a way that matters for investors. I want to separate what has been said by officials from what markets have actually done, because the two are not always aligned.

[30] Exact synthetic passage · span p2 · open on the writer's site

> First, the facts as I read them. The Federal Reserve has held its policy rate in a range that is high by the standards of the past decade, while the Bank of Japan has kept its short-term policy rate negative and continues to cap long-term yields through its yield curve control framework. That configuration has been the anchor for the yen carry trade, in which investors borrow in yen and invest in higher-yielding assets elsewhere. The carry trade has been profitable for a long time, but it is not without risk, as the sharp unwinding episodes of the past have shown.

[31] Exact synthetic passage · span p3 · open on the writer's site

> The Bank of Japan has repeatedly indicated that it won't hesitate to ease further if needed, but it has also acknowledged that the side effects of prolonged easing are becoming more visible. In its most recent statements, the bank has emphasized the importance of wage growth and inflation expectations in determining the future path of policy. Governor Ueda has said that the bank will carefully assess the data before making any changes, and that it does not target the exchange rate. That is a standard line, but it is worth taking at face value: the BoJ's mandate is price stability, not a particular yen level.

[32] Exact synthetic passage · span p4 · open on the writer's site

> At the same time, there have been hints that the bank is thinking about the sustainability of its framework. The yield curve control policy has been tweaked several times to allow long-term rates to move more flexibly, and each tweak has been interpreted by markets as a step toward normalization. But the bank has been careful to frame these moves as technical adjustments, not as a change in the overall stance of monetary policy. I think that is the right way to read them.

[33] Exact synthetic passage · span p5 · open on the writer's site

> Markets, however, have a habit of running ahead of central banks. The yen's recent appreciation has been driven in part by expectations that the rate differential will narrow, either because the Fed will cut rates or because the BoJ will raise them. But the actual differential remains wide. The 10-year US Treasury yield is still well above the 10-year JGB yield, even after the recent moves. That gap is the compensation for currency risk, and it is why the carry trade persists.

[34] Exact synthetic passage · span p6 · open on the writer's site

> I have seen some commentary suggesting that the yen's move is a sign that the carry trade is dead. I am not convinced. Carry trades have a tendency to revive when volatility subsides, and the underlying differential is still substantial. What has changed is that the market is now more alert to the risk of a policy shift. That means the carry trade is likely to be more volatile, with sharper unwinds when data surprises. For investors, that is a reason to size positions carefully and to consider hedging currency risk more actively.

[35] Exact synthetic passage · span p7 · open on the writer's site

> So what does this mean for portfolios? First, the yen's strength, if it persists, will affect the returns of Japanese assets for unhedged foreign investors. A stronger yen reduces the yen value of overseas earnings for Japanese exporters, which can weigh on equity prices. But it also makes Japanese assets cheaper for foreign investors when measured in their own currency. The net effect depends on the investor's base currency and hedging policy.

[36] Exact synthetic passage · span p8 · open on the writer's site

> Second, the rate differential is not just a yen story. It affects global bond markets, because Japanese investors are major buyers of foreign bonds. If the differential narrows, Japanese investors may repatriate capital, which could put upward pressure on JGB yields and downward pressure on US and European yields. That is a slow-moving process, but it is worth watching.

[37] Exact synthetic passage · span p9 · open on the writer's site

> Third, I would caution against extrapolating from recent price action. The yen has strengthened, but it is still weak by historical standards. The differential is still wide. The BoJ is still cautious. And the Fed is still data-dependent. In that environment, I prefer to focus on the range of possible outcomes rather than a single forecast. For investors, that means stress-testing portfolios for both a stronger yen and a weaker yen, and for both a narrowing and a widening differential.

[38] Exact synthetic passage · span p10 · open on the writer's site

> The yen's recent move is a reminder that currency markets can turn quickly, especially when they are positioned one way. The rate differential remains the dominant driver, but it is not the only one. I will be watching the data, the rhetoric from the BoJ and the Fed, and the positioning of carry traders. As always, I will try to separate what is said and what gets done. That distinction has served me well in the past, and I suspect it will continue to do so.

[39] Exact synthetic passage · span p11 · open on the writer's site

> For now, I am keeping an open mind. The yen could strengthen further if the BoJ signals a policy shift, or it could weaken if the Fed delays rate cuts. Either way, investors should be prepared for volatility. That is not a prediction; it is a recognition of uncertainty. And in my experience, acknowledging uncertainty is the first step toward managing it.

Kestrel Semiconductor exchange filing: TSMC capacity agreement

Open Records · or-kestrel-tsmc-filing-2026-09-29 · v1 · free
Receipt IDs: none
Attribution: Open Records

[40] Exact synthetic passage · span p1 · open on the writer's site

> On 29 September 2026 Kestrel Semiconductor announced a five-year agreement with TSMC for N3P wafer capacity, 24,000 wafers a month from Q3 2027, plus advanced-packaging slots. The disclosure was made in an exchange filing published before the opening of trading. Kestrel will pay a US$1.1 billion prepayment and gave no margin guidance. The previous contract covered 16,000 wafers a month.

[41] Exact synthetic passage · span p2 · open on the writer's site

> The filing states that the agreement runs for five years from the start of volume production. Kestrel said the prepayment is scheduled in three installments, with the first due within thirty days of signing. The company did not disclose the per-wafer price or the total contract value beyond the prepayment figure.

[42] Exact synthetic passage · span p3 · open on the writer's site

> According to the filing, the N3P wafers will be produced at TSMC's Fab 18 in Tainan. The first commercial deliveries are expected in the third quarter of 2027. Kestrel said the ramp will be phased, with the full 24,000 wafers a month reached by the second quarter of 2028. The advanced-packaging slots cover CoWoS and InFO processes, also at TSMC facilities in Taiwan.

[43] Exact synthetic passage · span p4 · open on the writer's site

> The previous contract, signed in 2023, covered 16,000 wafers a month on the N5 process. Kestrel said the new agreement replaces that contract upon its expiration in June 2027. The filing notes that the N3P process offers higher transistor density and lower power consumption than N5, but does not provide performance figures.

[44] Exact synthetic passage · span p5 · open on the writer's site

> Kestrel will pay a US$1.1 billion prepayment to TSMC. The filing states that the prepayment is non-refundable and will be credited against future wafer purchases over the five-year term. The first installment of US$400 million is due by 29 October 2026. The second and third installments of US$350 million each are due by 31 January 2027 and 30 April 2027, respectively.

[45] Exact synthetic passage · span p6 · open on the writer's site

> The company said the prepayment will be funded from existing cash and a new revolving credit facility. Kestrel reported cash and equivalents of US$2.3 billion as of 30 June 2026. The filing does not disclose the interest rate or covenants of the credit facility.

[46] Exact synthetic passage · span p7 · open on the writer's site

> Kestrel gave no margin guidance in the filing. The company said it will provide updated financial targets at its next earnings call, scheduled for 12 November 2026. The filing does not include revenue or earnings forecasts related to the agreement.

[47] Exact synthetic passage · span p8 · open on the writer's site

> The filing gives no guidance on how the agreement affects gross margin or wafer pricing, and Kestrel did not comment on analyst coverage.

[48] Exact synthetic passage · span p9 · open on the writer's site

> The filing was made under the exchange's rules for material contracts. Kestrel's shares closed at NT$487.50 on 28 September 2026, the day before the announcement. The Taiwan Stock Exchange requires listed companies to disclose contracts that may affect shareholder equity by more than 5 percent.

[49] Exact synthetic passage · span p10 · open on the writer's site

> TSMC declined to comment beyond confirming the agreement. The Bank of Japan's quarterly Tankan survey, released on 1 October 2026, will include capital expenditure plans for the semiconductor sector. Kestrel said it has not received any regulatory objections to the agreement.

[50] Exact synthetic passage · span p11 · open on the writer's site

> The filing states that the agreement includes a force majeure clause and a dispute resolution mechanism under Singapore law. Kestrel said it does not expect the agreement to have a material impact on its operations in the current fiscal year.

[51] Exact synthetic passage · span p12 · open on the writer's site

> Kestrel's previous exchange filing on 15 March 2026 disclosed a letter of intent with TSMC for N3P capacity. That filing did not specify volumes or prepayment amounts. The 29 September 2026 filing supersedes the letter of intent.

[52] Exact synthetic passage · span p13 · open on the writer's site

> On 2 August 2026, Kestrel filed its second-quarter results, reporting revenue of NT$18.4 billion and a gross margin of 34.2 percent. The company did not update those figures in the latest filing. Kestrel's next scheduled disclosure is its third-quarter results on 12 November 2026.

[53] Exact synthetic passage · span p14 · open on the writer's site

> The filing is available on the Taiwan Stock Exchange's Market Observation Post System under reference number 2026-09-29-001. Kestrel Semiconductor is listed on the Taiwan Stock Exchange under ticker 6415. TSMC is listed under ticker 2330.

Kestrel Semiconductor Exchange Filings – July 25, 2026

Open Records · open-kestrel-semiconductor-exchange-filings-july-25-2026 · v1 · free
Receipt IDs: none
Attribution: Open Records

[54] Exact synthetic passage · span p1 · open on the writer's site

> On July 25, 2026, Kestrel Semiconductor published its quarterly exchange filings with the Taiwan Stock Exchange and the Tokyo Stock Exchange. The filings, dated July 24, 2026, cover the period from April 1 to June 30, 2026. Kestrel reported net revenue of NT$142.3 billion for the quarter, a figure that appears in the consolidated income statement on page 12 of the Taiwan filing. The company also disclosed a gross margin of 48.7 percent, down from 51.2 percent in the same quarter of 2025. Operating income was NT$38.9 billion, and net income was NT$31.4 billion. These figures are stated in the filing without adjustment for foreign exchange fluctuations.

[55] Exact synthetic passage · span p2 · open on the writer's site

> The filings state that Kestrel's 3-nanometer wafer capacity reached 120,000 wafers per month as of June 30, 2026, up from 95,000 wafers per month at the end of March 2026. The company attributed the increase to the completion of a phase-two expansion at its Fab 18 in Tainan. According to the filing, 5-nanometer capacity remained flat at 150,000 wafers per month. The filing also notes that Kestrel's advanced packaging capacity, specifically its CoWoS line, stood at 35,000 wafers per month at quarter-end, compared to 28,000 wafers per month in the prior quarter. Kestrel stated that it expects CoWoS capacity to reach 45,000 wafers per month by December 31, 2026. No independent verification of these capacity figures was provided in the filing.

[56] Exact synthetic passage · span p3 · open on the writer's site

> The filings disclose that Kestrel's average selling price for 3-nanometer wafers increased by 4.2 percent quarter-over-quarter. The filing attributes this increase to a shift in product mix toward higher-priced variants. For 5-nanometer wafers, the average selling price decreased by 1.8 percent. The filing lists two customers that each accounted for more than 10 percent of Kestrel's net revenue in the quarter: Customer A represented 22 percent, and Customer B represented 14 percent. The identities of these customers are redacted in the public version of the filing. Kestrel also reported that its top ten customers accounted for 68 percent of net revenue, up from 65 percent in the prior quarter.

[57] Exact synthetic passage · span p4 · open on the writer's site

> Kestrel's filing states that capital expenditure for the quarter was NT$98.7 billion, with 70 percent allocated to advanced nodes and 30 percent to specialty technologies. The company reaffirmed its full-year 2026 capital expenditure guidance of NT$360 billion to NT$380 billion. For the third quarter of 2026, Kestrel provided revenue guidance of NT$148 billion to NT$154 billion, and gross margin guidance of 47.5 percent to 49.5 percent. The filing notes that these projections are forward-looking statements and are subject to risks described in the risk factors section on pages 45 to 52.

[58] Exact synthetic passage · span p5 · open on the writer's site

> The same day, the Bank of Japan released its summary of opinions from the June 2026 monetary policy meeting. The summary, dated July 25, 2026, includes a statement from one board member that read, "We are monitoring the impact of semiconductor supply chain adjustments on domestic capital investment." The Bank of Japan did not change its policy rate at that meeting, leaving it at 0.5 percent. The summary also noted that export volumes from Japan to Taiwan increased by 3.1 percent in the second quarter of 2026 compared to the same period in 2025, according to data from the Ministry of Finance. A separate report from the Taiwan Semiconductor Industry Association, dated July 23, 2026, stated that total wafer fab equipment spending in Taiwan rose 12 percent year-over-year in the first half of 2026.

[59] Exact synthetic passage · span p6 · open on the writer's site

> Kestrel's filing includes a section on litigation. As of June 30, 2026, the company was a defendant in three patent infringement lawsuits filed in the United States District Court for the Western District of Texas. The filing states that Kestrel believes it has meritorious defenses but cannot predict the outcome. The filing also discloses that Kestrel repurchased 2.1 million of its own shares during the quarter at an average price of NT$1,050 per share. The company's board of directors declared a cash dividend of NT$4.50 per share, payable on September 15, 2026, to shareholders of record as of August 20, 2026. These details appear on pages 78 to 80 of the Taiwan filing.

[60] Exact synthetic passage · span p7 · open on the writer's site

> The information in this article is drawn from Kestrel Semiconductor's quarterly exchange filings filed with the Taiwan Stock Exchange and the Tokyo Stock Exchange on July 24, 2026, and from the Bank of Japan's Summary of Opinions released on July 25, 2026. Additional data came from the Taiwan Semiconductor Industry Association report dated July 23, 2026, and the Ministry of Finance trade statistics for the second quarter of 2026. All figures are as stated in those documents. No independent audit or verification of Kestrel's internal figures has been performed by this desk. The filing documents are available on the respective exchange websites. This article is dated July 25, 2026.

BoJ Signals Future Rate Path: A Closer Look

Basis Points · basis-boj-signals-future-rate-path-a-closer-look · v1 · free
Receipt IDs: none
Attribution: Basis Points

[61] Exact synthetic passage · span p1 · open on the writer's site

> Speaking as an ex-central banker, I've learned to parse policy statements with a fine-toothed comb, separating what is said from what is done. The Bank of Japan's latest communication, released on 2026-05-10, is a case in point. The bank kept its policy rate unchanged, as widely expected, but its language on the future path of rates contained a nuance that deserves closer inspection. In this newsletter, I will walk through the statement, the press conference, and the market reaction, while being careful to distinguish between the bank's intentions and the market's interpretation.

[62] Exact synthetic passage · span p2 · open on the writer's site

> The statement began with the usual assessment of economic conditions, noting that Japan's economy is recovering moderately, albeit with some weakness in exports. Inflation, as measured by the consumer price index excluding fresh food, has been hovering around the 2 percent target, but the bank emphasized that it needs to see more sustainable wage growth before making any definitive moves. Governor Ueda's remarks in the press conference were particularly noteworthy. He said that the bank is not in a hurry to adjust policy, but that it will not hesitate to act if necessary. This is a subtle shift from previous language, which emphasized patience. The bank also updated its projections, showing slightly higher inflation for fiscal 2026 and 2027, but lower growth for 2026. These projections are not forecasts per se, but they inform the bank's reaction function.

[63] Exact synthetic passage · span p3 · open on the writer's site

> Markets, as they often do, reacted swiftly. The yen strengthened initially, then gave back some gains as traders digested the details. The 10-year JGB yield ticked up by about 2 basis points, a modest move. But the real action was in the swaps market, where the implied path for the policy rate shifted. Before the meeting, overnight index swaps were pricing in a roughly 40 percent chance of a hike by the end of 2026. After the meeting, that probability rose to around 60 percent. This suggests that market participants interpreted the bank's language as slightly more hawkish than expected. However, I would caution against over-reading a single meeting. The bank has repeatedly stressed that it will be data-dependent, and the next few months of wage and price data will be crucial.

[64] Exact synthetic passage · span p4 · open on the writer's site

> The BoJ's decision comes amid a complex global backdrop. The Federal Reserve has been holding rates steady, while the European Central Bank has signaled a potential cut later this year. In this environment, the BoJ remains an outlier, still maintaining negative interest rates and yield curve control. But the bank is slowly normalizing policy, and the question is how fast it will move. One key consideration is the impact on the banking sector. Japanese banks have been struggling with low margins, and a gradual rise in rates could help their profitability. However, a sharp increase could cause disruptions, especially for smaller institutions. The bank is also mindful of the yen's exchange rate, which has been volatile. A weaker yen boosts exporters but hurts households through higher import costs. The bank does not target the exchange rate, but it monitors it closely.

[65] Exact synthetic passage · span p5 · open on the writer's site

> Looking ahead, there are several key events that could shape the BoJ's next move. First, the shunto wage negotiations in spring 2027 will be critical. If wage growth accelerates, the bank may feel more confident about raising rates. Second, the consumption tax hike scheduled for October 2026 could dampen consumer spending, which might make the bank more cautious. Third, global economic conditions, particularly in China and the US, will influence Japan's export sector. Finally, the bank's own communication will be key. Governor Ueda has been careful to avoid surprising markets, but he has also shown a willingness to adjust policy when needed. I would expect the bank to continue with a gradual approach, possibly raising rates once or twice by the end of 2027, but this is highly uncertain. As always, I will be watching the data and the bank's language closely.

[66] Exact synthetic passage · span p6 · open on the writer's site

> In conclusion, the BoJ's latest meeting did not bring any policy changes, but it did offer a subtle shift in tone. The bank is slowly moving toward normalization, but it is doing so cautiously. Markets have priced in a higher probability of a rate hike, but they may be getting ahead of themselves. The bank's decisions will depend on a range of factors, including wage growth, inflation, and global conditions. As an ex-central-bank economist, I'm wary of overinterpreting any single communication. The BoJ has a long history of surprising markets, and it is likely to continue doing so. For now, I will keep a close eye on the data and the bank's next move. This newsletter is for informational purposes only and should not be considered investment advice. I have no positions in any securities mentioned. The views expressed are my own and do not necessarily reflect those of any institution. I would like to thank my readers for their continued support. If you have any questions or comments, please feel free to reach out. Until next time, stay informed and stay cautious.

What this year's corporate green power round actually bought

Load Factor · lf-corporate-green-power-round-2026 · v1 · free
Receipt IDs: none
Attribution: Load Factor

[67] Exact synthetic passage · span p1 · open on the writer's site

> Every few months someone sends me a press release about a corporate green power deal and asks whether the factory in question now runs on sunshine. It almost never does, and the gap between the headline megawatts and the energy a plant actually uses is the most useful piece of arithmetic in this whole debate. This year's corporate green power round in Peninsular Malaysia closed its last contracts in September, so it is a good moment to do that arithmetic properly.

[68] Exact synthetic passage · span p2 · open on the writer's site

> The programme is simple in outline. A developer builds a farm, the grid carries the output, and a corporate buyer signs a long contract to take the energy at an agreed price, settled through the utility. The buyer never sees an electron from the farm. What it buys is a claim on a certain number of megawatt-hours a year, matched on paper against its own consumption, plus a certificate trail its auditors will accept.

[69] Exact synthetic passage · span p3 · open on the writer's site

> This round allocated a little over 600 MW of new capacity across roughly two dozen buyers. Most contracts run for 21 years, which matches the economic life the developers assume for their panels. The buyers are a predictable mix: data-centre operators in Johor, two glove makers, a brewery, several electronics plants in Kulim and Perak, and one cement works that I suspect signed mainly for the press coverage.

[70] Exact synthetic passage · span p4 · open on the writer's site

> Here is the arithmetic I wish every press release included. A farm in the peninsula runs at a capacity factor of roughly 0.15 to 0.17. That means a 10 MW contract delivers, on average, about 1.2 GWh a month, or about 14,000 MWh a year at the midpoint. A factory with a steady 10 MW load uses about 87,600 MWh a year. So a contract with the same nameplate as the factory's load covers about one sixth of its energy. Not all of it. Not half of it. One sixth.

[71] Exact synthetic passage · span p5 · open on the writer's site

> That ratio is the thing to hold on to, because it applies to every buyer on the list. When a company says it has contracted capacity equal to its peak demand, it has bought perhaps a sixth of its annual electricity. When it has contracted a fraction of its peak, the share falls accordingly, and the press release rarely says so.

[72] Exact synthetic passage · span solar-contract · open on the writer's site

> The award that has generated the most email is the semiconductor one. Kestrel Semiconductor signed a 21-year contract on 8 September 2026 for 40 MW of solar capacity, allocated to its Penang Phase 2 advanced-packaging plant. I have written about that plant's grid connection before, so I already had its load profile to hand.

[73] Exact synthetic passage · span solar-share · open on the writer's site

> The arithmetic goes like this. Forty megawatts at a capacity factor of 0.16 is about 56 GWh a year. The plant's average draw should settle near 35 MW once the lines are fully loaded, which puts its consumption somewhere between 300 and 335 GWh a year. Divide one by the other and the contract covers about 17% of the Phase 2 plant's expected annual electricity use. That is a respectable share for a single contract, and roughly what I would have advised. It is not a plant running on sunshine, and to its credit nobody at the company has claimed that it is.

[74] Exact synthetic passage · span p8 · open on the writer's site

> There is a second reason the share is modest, and it is about timing rather than totals. A packaging line runs around the clock, and some of its hardest-working shifts are at night, when the farm produces nothing. On paper the contract is matched over a year. Hour by hour, a plant like this still draws almost all of its night-time load from gas-fired generation on the mainland. If a buyer later wants hourly matching rather than annual matching, it will need storage, and storage at that scale costs real money.

[75] Exact synthetic passage · span p9 · open on the writer's site

> Three things. First, whether the next round raises the cap on contract size, which would let large industrial buyers move well past one fifth of their consumption. Second, whether the utility publishes settlement data for these contracts, so that delivered energy can be compared with promised energy. Third, whether buyers start asking for shorter contracts with price reviews, because 21 years is a long time to fix a price for a technology whose costs are still falling.

[76] Exact synthetic passage · span p10 · open on the writer's site

> For now, the round is a useful step, and the press releases are a useful reminder that megawatts and megawatt-hours are two different units. Read them that way and most of the excitement, and most of the cynicism, goes away.

Cooling Load Dynamics in Southeast Asian Data Halls

Load Factor · load-cooling-load-dynamics-in-southeast-asian-data-halls · v1 · free
Receipt IDs: none
Attribution: Load Factor

[77] Exact synthetic passage · span p1 · open on the writer's site

> Last month I spent three days in a colocation facility on the outskirts of Singapore, watching a 4.2 MW data hall breathe. The operators were proud of their power usage effectiveness, a number they quote to prospective tenants the way a hotel quotes its star rating. But what struck me was not the PUE. It was the water.

[78] Exact synthetic passage · span p2 · open on the writer's site

> In temperate climates, cooling a data hall is mostly a sensible-heat problem. You move air, you reject heat, you occasionally run a chiller. In the tropics, it is a latent-heat problem wearing a sensible-heat costume. Every cubic metre of outside air you admit carries water vapour, and that vapour must be condensed out before the air can do useful work. Condensation is not free. It is, in fact, the single largest hidden load in most Southeast Asian facilities I have audited.

[79] Exact synthetic passage · span p3 · open on the writer's site

> Let me show you the arithmetic, because the arithmetic is where the argument lives.

[80] Exact synthetic passage · span p4 · open on the writer's site

> Take a hall with an IT load of 4.2 MW. Suppose the operators run a conventional chilled-water plant with a design PUE of 1.45. That gives a total facility load of 6.09 MW, of which 1.89 MW is cooling and losses. Now suppose they decide to economise by admitting outside air at 32 degrees Celsius and 80 percent relative humidity, which is a typical afternoon in Kuala Lumpur or Jakarta.

[81] Exact synthetic passage · span p5 · open on the writer's site

> At those conditions, the air holds roughly 24 grams of water per kilogram of dry air. To bring it to a supply condition of 13 degrees Celsius and 95 percent relative humidity, you must reduce that to about 9 grams per kilogram. That is 15 grams of water condensed for every kilogram of dry air you push through the hall.

[82] Exact synthetic passage · span p6 · open on the writer's site

> A 4.2 MW hall typically needs around 200 cubic metres per second of supply air, which is about 240 kilograms of dry air per second. Multiply by 15 grams and you get 3.6 kilograms of condensate per second. That is 3.6 litres per second, or about 311,000 litres per day. Three hundred and eleven cubic metres of water, every day, just to keep the air dry enough to cool the servers.

[83] Exact synthetic passage · span p7 · open on the writer's site

> That water does not appear in the PUE. It appears in the water bill, and increasingly in the sustainability report, but it is invisible to the metric that the industry uses to compare facilities. This is a problem of measurement, not just of engineering.

[84] Exact synthetic passage · span p8 · open on the writer's site

> The standard playbook for efficient data centres was written in Oregon and Ireland. It assumes cool, dry outside air for much of the year. It assumes free cooling hours in the thousands. It assumes that water is cheap and abundant, or at least that its cost is second-order.

[85] Exact synthetic passage · span p9 · open on the writer's site

> In Southeast Asia, none of those assumptions hold. Free cooling hours are close to zero in the lowlands. Water is a political resource, not just a utility. And the latent load is not a seasonal nuisance; it is the baseline.

[86] Exact synthetic passage · span p10 · open on the writer's site

> I have seen designs imported wholesale from European reference architectures that specify indirect evaporative cooling, only to discover that the wet-bulb temperature in April is 28 degrees Celsius. Evaporative cooling cannot reach a supply temperature below the wet-bulb, so the design is dead on arrival. The fix is mechanical refrigeration, which means more electricity, which means a higher PUE, which means the facility looks worse on a scorecard that was never designed for this climate.

[87] Exact synthetic passage · span p11 · open on the writer's site

> This is not a failure of engineering. It is a failure of benchmarking. We are comparing apples to durians and pretending the difference is a matter of skill.

[88] Exact synthetic passage · span p12 · open on the writer's site

> I want to tread carefully here, because there is a lot of hand-waving in this space. Let me cite what I can verify.

[89] Exact synthetic passage · span p13 · open on the writer's site

> The International Energy Agency's 2024 report on data centres and networks estimates that data centres consumed about 460 terawatt-hours globally in 2022, roughly 1.5 percent of total electricity demand. That number is contested, and the IEA itself notes the uncertainty. But the direction is not contested: demand is rising, and it is rising fastest in regions where cooling is hardest.

[90] Exact synthetic passage · span p14 · open on the writer's site

> Closer to home, Singapore's Energy Market Authority reported in its 2023 Singapore Energy Statistics that the information and communications sector accounted for about 7 percent of the country's electricity consumption, with data centres making up the bulk of that. The government has since imposed a pause on new data centre construction, not because of electricity alone, but because of water and land.

[91] Exact synthetic passage · span p15 · open on the writer's site

> TSMC's 2023 sustainability report offers a useful comparison from a different industry. The company reported total water withdrawal of about 220 million cubic metres in 2022, with a recycling rate above 85 percent. That is a semiconductor fab, not a data hall, but the scale is instructive. A single large data centre campus in the tropics can withdraw tens of millions of litres per year for cooling, and unlike TSMC, most operators have not yet built the recycling infrastructure to match.

[92] Exact synthetic passage · span p16 · open on the writer's site

> The Bank of Japan's 2024 Financial System Report included a chapter on climate-related risks to financial stability, noting that water stress in Southeast Asia poses a material risk to the region's industrial output. The BoJ is not an energy regulator, but its analysts are paying attention to the same physics I am describing.

[93] Exact synthetic passage · span p17 · open on the writer's site

> The solutions are not exotic. They are, in fact, boring, which is why they are often overlooked.

[94] Exact synthetic passage · span p18 · open on the writer's site

> First, stop admitting outside air when you do not need to. A sealed hall with a dedicated dehumidification loop can cut latent load by half or more. The trade-off is that you must handle internal moisture sources, which means better sealing and better monitoring. But the arithmetic favours sealing in almost every tropical case I have modelled.

[95] Exact synthetic passage · span p19 · open on the writer's site

> Second, raise the supply temperature. Every degree you raise the chilled-water supply temperature reduces the lift on the compressor and improves the chiller's coefficient of performance. Modern servers can tolerate 27 degrees Celsius inlet air, and some can go higher. The ASHRAE TC 9.9 guidelines have been moving in this direction for a decade. The industry has been slow to follow.

[96] Exact synthetic passage · span p20 · open on the writer's site

> Third, measure water separately from electricity. A PUE that ignores water is a half-truth. I would like to see a water usage effectiveness metric reported alongside PUE, with the same prominence. Some operators already do this. Most do not.

[97] Exact synthetic passage · span p21 · open on the writer's site

> None of this is glamorous. It will not win a design award. But it will keep the lights on, the servers cool, and the water in the reservoir where it belongs. That seems like a reasonable trade to me.

Malaysia Data-Centre Approvals – June 20, 2026

Open Records · open-malaysia-data-centre-approvals-june-20-2026 · v1 · free
Receipt IDs: none
Attribution: Open Records

[98] Exact synthetic passage · span p1 · open on the writer's site

> On 20 June 2026, Malaysia's Investment, Trade and Industry Ministry (MITI) published its monthly data-centre project register for May 2026. The register lists 14 approved data-centre projects with a combined committed investment of RM 9.4 billion. Of these, nine are located in Johor, three in Selangor, one in Penang and one in Negeri Sembilan. MITI states that the approvals cover a combined 412 megawatts of contracted electrical capacity.

[99] Exact synthetic passage · span p2 · open on the writer's site

> The same register records that two previously approved projects in Johor were withdrawn by their developers in May 2026, citing grid connection timelines. MITI does not name the developers. The register states that withdrawn capacity totalled 96 megawatts.

[100] Exact synthetic passage · span p3 · open on the writer's site

> Tenaga Nasional Berhad (TNB) published its May 2026 grid connection queue on 18 June 2026. The queue lists 31 data-centre connection applications under study, with a combined requested capacity of 2.7 gigawatts. TNB states that 11 of the 31 applications are in Johor, 9 in Selangor, 5 in Penang, 4 in Negeri Sembilan and 2 in Kedah. TNB states that the average study period for applications received in the first quarter of 2026 was 214 days.

[101] Exact synthetic passage · span p4 · open on the writer's site

> The Energy Commission (Suruhanjaya Tenaga) issued a statement on 12 June 2026 confirming that it had received 22 applications for electricity supply licences for data-centre operations between January and May 2026. The Commission states that 8 licences were issued in the period, 6 were refused and 8 remain under review. The Commission does not publish applicant names.

[102] Exact synthetic passage · span p5 · open on the writer's site

> The National Water Services Commission (SPAN) stated on 9 June 2026 that it had received 17 applications for water supply commitments for data-centre projects in the first five months of 2026. SPAN states that 10 applications were approved, 4 were deferred pending further documentation and 3 were rejected. SPAN states that approved commitments total 38 million litres per day.

[103] Exact synthetic passage · span p6 · open on the writer's site

> The Johor state government published a statement on 5 June 2026 stating that it had received 26 data-centre planning permission applications in the first five months of 2026 and had approved 15. The statement says the state has set a review threshold of 20 megawatts for projects requiring state-level technical review.

[104] Exact synthetic passage · span p7 · open on the writer's site

> TSMC filed a notification with the Malaysia Investment Development Authority (MIDA) on 3 June 2026 regarding a planned advanced packaging facility in Penang. The filing states that the facility is intended to support data-centre accelerator packaging and that construction is scheduled to begin in the first quarter of 2027. The filing does not state a capacity figure. MIDA confirmed receipt of the filing on 4 June 2026.

[105] Exact synthetic passage · span p8 · open on the writer's site

> A joint venture registered in Singapore, identified in filings as Horizon Compute Holdings Pte Ltd, filed with the Companies Commission of Malaysia on 28 May 2026 to establish a Malaysian subsidiary. The filing lists a paid-up capital of RM 250 million and states the subsidiary's intended activity as data-centre development and operation. The filing does not name a site.

[106] Exact synthetic passage · span p9 · open on the writer's site

> YTL Power International Berhad stated in a filing dated 11 June 2026 that its data-centre campus in Kulai, Johor, had reached 120 megawatts of operational capacity. The filing states that a further 80 megawatts is under construction and scheduled for completion in the fourth quarter of 2026.

[107] Exact synthetic passage · span p10 · open on the writer's site

> GDS Holdings stated in a press release dated 16 June 2026 that it had signed a power purchase agreement with TNB for a 64-megawatt facility in Selangor. The release states that the agreement runs for 15 years and that commercial operation is targeted for the second quarter of 2027.

[108] Exact synthetic passage · span p11 · open on the writer's site

> The Bank of Japan (BoJ) published its monetary policy statement on 17 June 2026, holding its policy rate unchanged. The statement does not reference Malaysia or data-centre investment. The BoJ's next scheduled policy meeting is on 30 and 31 July 2026.

[109] Exact synthetic passage · span p12 · open on the writer's site

> Malaysia's Department of Statistics released May 2026 trade data on 19 June 2026. The data record a 14.2 per cent year-on-year increase in exports of electrical and electronic products, which the Department attributes in part to semiconductor and data-centre equipment shipments. The Department does not publish a separate data-centre equipment category.

[110] Exact synthetic passage · span p13 · open on the writer's site

> Bank Negara Malaysia published its international reserves statement on 13 June 2026, recording reserves of USD 118.4 billion as at 31 May 2026. The statement does not reference data-centre investment.

[111] Exact synthetic passage · span p14 · open on the writer's site

> The ASEAN Secretariat published a note on 10 June 2026 stating that member states had agreed to a common reporting template for cross-border data-centre investment statistics. The note states that the template will take effect on 1 January 2027 and that reporting will be voluntary in the first year.

[112] Exact synthetic passage · span p15 · open on the writer's site

> MITI's register states that 6 data-centre applications received in May 2026 remain under review and that decisions are expected by 31 July 2026. TNB states that it will publish its June 2026 connection queue on 18 July 2026. SPAN states that its next water commitment review is scheduled for 14 July 2026. The Johor state government states that its next planning permission decisions are scheduled for 3 July 2026. The Energy Commission states that its next licence decisions are scheduled for 24 July 2026. No further statements were recorded as of 20 June 2026.

Kestrel Semiconductor advanced-packaging lead times in Malaysia: the dated series

The Fab Floor · fab-floor-kestrel-penang-lead-times · v1 · bought / granted
Receipt IDs: x402-3081E9C26915BDEA7A4FB9154B27C6E1C9F597121BAEE5F6967D8C63CD256365
Attribution: The Fab Floor

[113] Exact synthetic passage · span p1 · open on the writer's site

> I run procurement for a mid-size fabless shop, and I have spent most of 2026 staring at Kestrel Semiconductor's Penang advanced-packaging queue. Here is the blunt version: the queue has been shrinking, and it has been shrinking faster than anyone on the buy side expected. I have the dated numbers, and I have the week counts, because that is the only way to talk about this business without getting lied to.

[114] Exact synthetic passage · span p2 · open on the writer's site

> On 1 June 2026, Kestrel's Penang advanced-packaging lead time was 26 weeks. That was the number we built our H2 allocation plan around. It was ugly. It meant anything we booked in June landed in December, which meant our Q4 customer commitments were effectively locked before we had a single wafer out of the foundry. We were quoting our own customers at 30 weeks and padding the back end because we did not trust the 26.

[115] Exact synthetic passage · span lead-times · open on the writer's site

> By 1 August 2026, the same Penang line was quoting 22 weeks. That is a four-week improvement in eight weeks, which sounds modest until you annualise it. Then on 15 September 2026, the number came in at 18 weeks. So from 1 June 2026 to 15 September 2026, we went from 26 weeks to 18 weeks, a cut of 31%, while capacity grew from 11,000 units to 16,000 units a month. That is eight weeks of lead time gone in fifteen weeks of calendar time. I have been doing this for eleven years and I have never seen a packaging queue compress that fast without a demand shock behind it.

[116] Exact synthetic passage · span p4 · open on the writer's site

> The obvious explanation is capacity. Kestrel commissioned Phase 2 at Penang on 15 August 2026. That is the date I have from the supplier's own capacity notice, and it lines up with the 1 August 2026 quote of 22 weeks, because the line was already loading Phase 2 tools in late July. Phase 2 is the advanced-packaging expansion, the one they have been teasing since 2024. It is real, it is running, and it is the reason the queue is moving.

[117] Exact synthetic passage · span p5 · open on the writer's site

> The less obvious explanation is mix. Kestrel has been quietly pushing lower-layer flip-chip work off Penang and onto its other back-end sites, which frees the Penang line for the high-layer-count stuff that actually pays. I do not have that in writing. I have it from three separate conversations with people who would know, and the behaviour of the quotes supports it. Our simple parts are suddenly quoting out of a different site. Our hard parts are still Penang, but they are quoting shorter.

[118] Exact synthetic passage · span p6 · open on the writer's site

> Here is the part that matters for anyone building a dual-source strategy. The Johor alternative, the one everyone has been treating as the safe fallback, was still at 24 weeks on 15 September 2026. Read that again. Penang is at 18 weeks. Johor is at 24 weeks. The fallback is now six weeks slower than the primary.

[119] Exact synthetic passage · span p7 · open on the writer's site

> That is a complete inversion of where we were in the spring. Back in Q1, Johor was the fast lane and Penang was the queue you endured because you had no choice. Now the fast lane is Penang and Johor is the queue. If you are still splitting your volume 50/50 on the assumption that Johor is the safer bet, you are paying a six-week penalty for the privilege of being wrong.

[120] Exact synthetic passage · span p8 · open on the writer's site

> I am not saying abandon Johor. I am saying reprice it. If your allocation model still treats Johor as the low-risk path, your model is stale. The risk premium has flipped.

[121] Exact synthetic passage · span p9 · open on the writer's site

> I moved our Q4 volume. Not all of it, but the majority. We had been running roughly 60/40 Penang/Johor since the spring, and I have pushed that to 80/20 as of the third week of September. The reason is simple arithmetic. At 18 weeks, Penang lands our Q4 build in the first half of January. At 24 weeks, Johor lands it in early March. One of those keeps our customer commitments. The other one does not.

[122] Exact synthetic passage · span p10 · open on the writer's site

> The counterargument is concentration risk. If Penang sneezes, we are exposed. That is fair, and I am not pretending it is not. But concentration risk is a probability, and a six-week lead-time gap is a certainty. I will take the certain six weeks and manage the probability with buffer stock on the two parts where a Penang hiccup would actually hurt us.

[123] Exact synthetic passage · span p11 · open on the writer's site

> The other thing I am doing is re-quoting everything. Every PO we have out with Kestrel, I have asked for a fresh lead time against the 15 September 2026 number. Some of them came back at 18 weeks. Some came back longer because of specific layer counts or specific substrate constraints. The headline number is not the number for your part. The headline number is the number for the easiest part on the line. Do not plan against the headline.

[124] Exact synthetic passage · span p12 · open on the writer's site

> Kestrel is not doing this out of generosity. They are doing it because Phase 2 gave them capacity and because they want the high-layer-count volume, and the way you win high-layer-count volume is by being the fastest quote in the room. The 26 weeks to 18 weeks move is a commercial play as much as a capacity play. That is fine. I do not need my suppliers to be charitable. I need them to be predictable, and right now Penang is more predictable than it has been all year.

[125] Exact synthetic passage · span p13 · open on the writer's site

> The question is whether it holds. Phase 2 is commissioned, but commissioning is not the same as mature yield. If the new line runs into a substrate problem in October, that 18 weeks goes back to 22 in a hurry. I have seen it happen. So I am treating the 18 weeks as real but not permanent. I am booking against it. I am not signing annual contracts against it.

[126] Exact synthetic passage · span p14 · open on the writer's site

> If you are still quoting your own customers off a 26-week assumption, you are leaving weeks on the table. If you are still routing volume to Johor because it used to be the fast lane, you are paying for a memory. The numbers as of 15 September 2026 are 18 weeks Penang, 24 weeks Johor, and a Phase 2 line that came online on 15 August 2026. That is the whole picture. Act accordingly.